

## Lecture 27

3/30/26  
JRA

Exam due Friday 5pm - submit to Moodle  
If you have questions, please ask.

See email w/ notes from Friday.

US sims - 2 visible

2 to be made visible

---

How do you create a 2D image?



Need to <sup>send pulse &</sup> get echo  
along more than 1D.

Two common ways to get 2D image.

In past, w/ one-element transducer, had  
to scan by hand

Now can have 256 or 512 elements  
Send & detect a few at a time

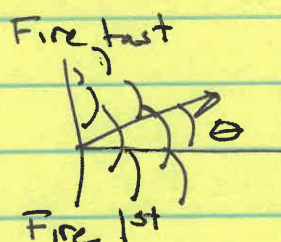
Carotid  
imaging

Linear transducer -

Phased array transducer - sweep | <sup>see</sup> animation

It's possible to steer beam by manipulating the firing pattern

Heat  
imaging



Fire farthest elements first, so they travel farther

Remaining HW is all sim based

## Image Quality

1. Did contrast first  
Contrast - DONE

Comes from echo strength

Big echoes = white

Weaker = shades of gray

No echo = black

Can characterize strength of echo for specular reflection (which occurs from smooth surface)

$$R = \frac{(z_1 - z_2)^2}{(z_1 + z_2)^2}$$

$R = 1 \text{ to } 2\%$  w/ soft tissues

Ideal because it leaves plenty of US for deeper structures

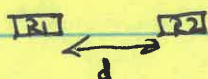
## 2) Resolution

Just as w/ light microscopy, there is both lateral & axial resolution

### (a) Lateral Resolution

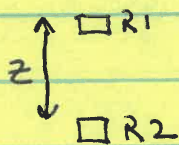
↓ Beam propagation (a pulse prop)

$R_1, R_2 \perp$   
to direction  
of beam



) ( Call pulse  
'beam' because  
it has a profile

### (b) Axial Resolution



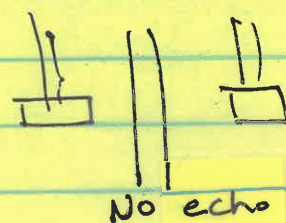
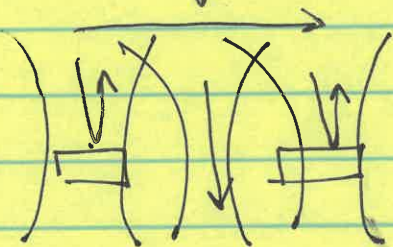
## Resolution Benchmark

Two reflectors are laterally resolvable if the beam width,  $BW$ , is less than the lateral separation.

$$BW < d$$

## Lateral Resolution

Get echo from  $R_1$ , echo from  $R_2$ , need an echo free region in between.

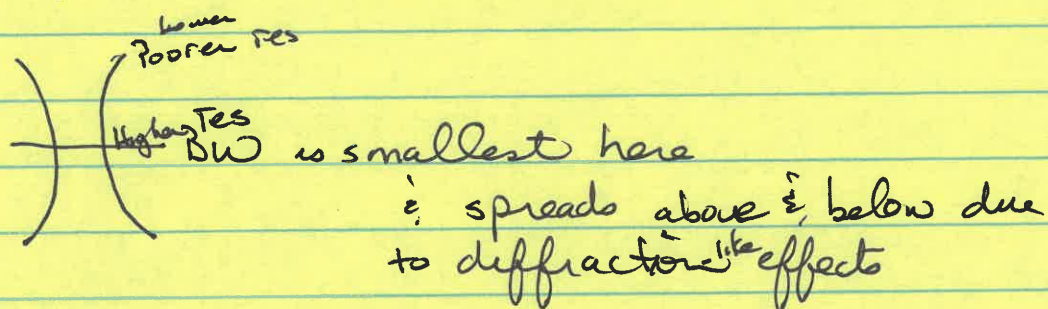


Argument is similar to argument for laser scanning methods

If beam width  $> d$  will get an echo even if centered over gap because some beam will bounce off reflectors

You can focus US just like a laser beam

One option is an acoustic lens



Try to focus beam at level of reflectors of interest

Dr focuses at depth where he wants resolution at best

What about frequency? Choice 1-20 MHz

Just as w/ microscopy,

Resolution improves as  $\lambda \uparrow$   $\epsilon$   $\lambda \downarrow$

BW at focus

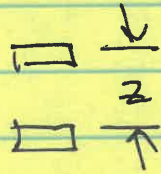
$$BW = \frac{\lambda F}{A}$$

$\lambda$  - focal length of lens  
A - aperture size of ap creating US

Want res  $\approx$  mm

Fire multiple elements simultaneously to increase A

Axial

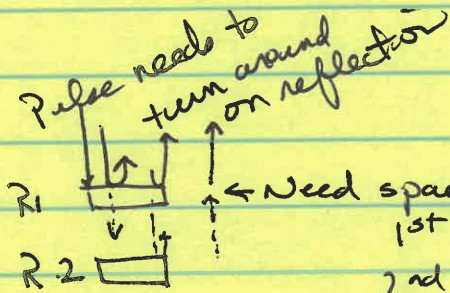
Label  
on drawing  
above
 $z$  = separation  
between two  
reflectors along  
direction of  
beam propagation

What dictates axial resolution

Punch Line: need an echo free region

spatial pulse length

$SPL < 2z$

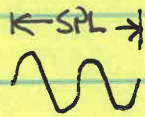


Need space between two pulses

1st reflect

2nd reflect

Now take pulse View of US

Typically  $SPL = 2-3\lambda$ Why is  $SPL < 2z$  the resolution condition?

If tip of 2nd reflected catches tail of 1st reflected then NO resolution

Echo overlap  $\Rightarrow$  NO resolutionThis will happen if  $SPL \geq 2z$

$$SPL \approx 3\lambda$$

Want a small SPL for good resolution

Beam diam small  $\Rightarrow$  good lateral resolution  
 SPL small  $\Rightarrow$  good axial resolution

Want small  $\lambda \Rightarrow$  High frequency

Big punchline - to <sup>get the best</sup> maximize lateral  
 & axial resolution  
 use high freq / low  $\lambda$  US

Why even use lower-freq US?

Q Why do Drs often/sometimes use lower  $\lambda$  US?

A Lower  $\lambda$  penetrates better

Energy in U beam quickly gets dissipated into heat energy. Then no energy in US beam is done.

Faster motion due to higher freq, energy is dissipated faster

Energy dissipation is faster at higher  $\nu$

If you need to go deep, use lower  $\nu$

People think about half depth  $L_{1/2}$

$L_{1/2}$  is depth in cm over which half the energy is lost

Handy rule of thumb

$$\text{Soft tissues } L_{1/2} \approx \frac{6 \text{ cm}}{\nu} \quad \nu \text{ in MHz}$$

$$\nu = 1 \text{ MHz} \quad L_{1/2} \Big|_{\nu=1 \text{ MHz}} = 6 \text{ cm}$$

$$L_{1/2} \Big|_{\nu=10 \text{ MHz}} = 0.6 \text{ cm}$$

Each  $L_{1/2}$  is  $1/2$  less energy

Is reflection weak because of attenuation  
or poor reflector

High  $\nu$  that maxes res also maxes attenuation

There is a way to compensate for attenuation

Med imaging companies make phantoms for testing & learning

You can use time gain compensation TGC to correct for attenuation  
Base correction on time

Doppler US - measure BFlow

Can get sampling issues

---

© 2026 Beth A. Scalettar & James R. Abney